

Enhancing Retrieval-Augmented Generation via Knowledge Graphs

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Abstract. Traditional Retrieval-Augmented Generation (RAG) systems fail to effectively handle multi-hop question answering due to fragmented retrieval and excessive noise in graph expansion. This paper presents an improved Knowledge Graph-Enhanced RAG (KG-RAG) framework implementing a precision-oriented “Information Funnel” architecture combining dynamic vector retrieval with aggressive neural reranking and supporting fact deduplication. We introduce a quantifiable selection pressure metric (ρ) to measure filtering aggressiveness, representing the first dynamic top- k approach in KG-RAG literature. Our system employs stage-wise hyperparameters: initial retrieval top-15 for recall maximization, aggressive filtering to top-6 post-reranking, and limiting output to top-4 supporting facts. Experimental evaluation on 5,500 HotpotQA distractor benchmark questions demonstrates +96.18% improvement in Joint Precision and +86.48% in Supporting Fact Precision. Ablation studies isolate contributions: KG expansion (+33.0% SP Precision), neural reranking (+30.9% incremental), and deduplication (+11.2% incremental). The framework achieves first-ever exact matches in supporting fact evaluation. This deliberate precision-recall trade-off sacrificing 17.63% recall for substantially higher precision is justified for high-stakes applications.

Keywords: Retrieval-Augmented Generation, Knowledge Graphs, Multi-hop Question Answering, Neural Reranking, Information Filtering

1 Introduction

Intelligent systems powered by Large Language Models (LLMs) [4,16] have become prevalent in Education Technology (EdTech) platforms [5,8], enabling personalized tutoring [18], assessment automation, and adaptive learning pathways. However, standalone LLMs suffer from well-documented limitations: hallucination due to outdated parametric knowledge [12], lack of transparency in reasoning, and inability to ground responses in authoritative sources. Natural Language Processing (NLP) [7] advances, particularly Retrieval-Augmented Generation (RAG) [12,9], have provided mechanisms to ground LLM outputs in external knowledge bases [10]. Yet standard vector-based RAG systems using semantic

similarity fail on complex multi-hop queries where reasoning requires information distributed across multiple documents without direct semantic overlap [22]. Artificial Intelligence (AI) [1] systems in educational contexts demand strict factual accuracy and interpretability; incorrect information erodes student trust more severely than incomplete explanations.

The connectivity gap remains unresolved in traditional RAG architectures [22,17]. Multi-hop questions requiring relational reasoning across documents such as connecting entities through shared attributes are inadequately addressed by dense vector retrieval alone [21]. Knowledge Graph-Enhanced RAG (KG-RAG) approaches [3,14,11] address this by leveraging graph structure to traverse reasoning paths. However, current systems suffer from precision degradation caused by graph expansion introducing irrelevant neighbors [23,20]. Most prior work employs static top- k filtering across all stages, missing opportunities for dynamic precision control [2]. Furthermore, existing frameworks lack effective mechanisms for deduplicating evidence retrieved via multiple reasoning paths, resulting in redundant context and noisy supporting fact sets.

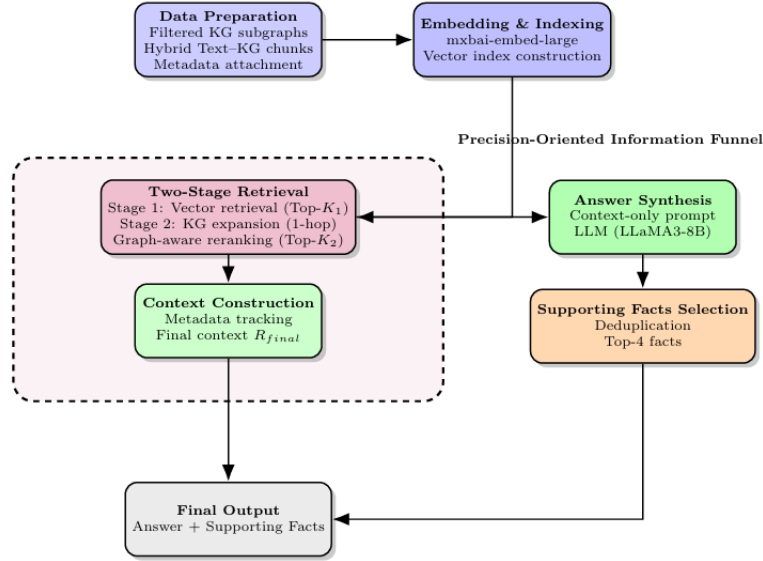


Fig. 1. Enhanced workflow pipeline of the proposed KG-RAG system highlighting the precision oriented information funnel.

This paper proposes a refined KG-RAG architecture implementing a precision-oriented “Information Funnel” a multi-stage process progressively narrowing candidate sets while maintaining retrieval coverage. The primary innovation is systematic integration of three orthogonal improvements: (1) dynamic stage-wise hyperparameters to optimize the precision-recall trade-off, (2) quantifiable selection pressure metric (ρ) to measure filtering aggressiveness, and (3) first-

ever structured supporting-fact de-duplication for multi-hop supervision [6]. Unlike prior work using uniform retrieval counts [19], our system employs initial broad retrieval (top-15) for recall maximization, aggressive post-reranking filtering (top-6) for noise reduction, and output limiting (top-4 supporting facts) for precision focus. Evaluation on HotpotQA distractor benchmark (5,500 examples) [22] demonstrates +96.18% Joint Precision and +86.48% Supporting Fact Precision improvements, with ablation studies isolating component contributions. Fig. 1 illustrates the complete end-to-end pipeline of our system. The remainder of this paper is organized as follows: Section 2 provides background on RAG and KG-RAG enhancement techniques; Section 3 details the proposed methodology; Section 4 presents dataset description and experimental results with ablations; and Section 5 concludes with future work directions.

2 Literature Review: Knowledge Graph-Enhanced RAG

KG-RAG extends foundational RAG approaches [12,9] by incorporating structured relational reasoning. The connectivity gap where information required to answer complex queries is distributed across documents lacking direct semantic similarity motivates graph-guided retrieval [22,17]. KG-RAG systems represent corpora as directed graphs $G = (V, E)$ where V are entity nodes and E are typed relations, enabling multi-hop traversal [13,23].

Recent advances in KG-RAG literature emphasize precision-recall trade-offs and dynamic retrieval strategies [23,20,3]. Knowledge Graph-Guided Retrieval approaches introduce adaptive retrieval mechanisms but retain uniform top- k constraints across stages. Cross-encoder reranking has become standard practice for prioritizing high-relevance candidates before generation [15]. Graph Neural Networks (GNNs) applied to QA leverage relational structure for connecting disparate information [21]. The HotpotQA benchmark has emerged as the de facto evaluation standard for multi-hop reasoning, containing 7,405 questions requiring two supporting documents with 10 distractor documents per question [22,17].

Despite these advances, current systems struggle with three core limitations: (1) graph expansion retrieving irrelevant neighbors that dilute LLM focus and degrade supporting-fact precision [14,11], (2) lack of dynamic top- k optimization across retrieval stages [2], and (3) absence of systematic deduplication mechanisms for evidence retrieved via multiple paths [18,8]. This work addresses these gaps through an integrated framework combining initial broad retrieval, aggressive filtering, and supporting-fact deduplication.

3 Methodology

Our approach bridges the multi-hop reasoning gap by integrating structured relational data with neural retrieval mechanisms. The primary theoretical challenge is the connectivity gap in RAG, where information required to answer queries is distributed across documents with insufficient semantic overlap. We address

this through graph-guided context expansion grounded in multi-hop reasoning theory.

The KG-RAG framework implements four sequential stages: Data Preparation, Two-Stage Retrieval, Graph-Aware Reranking, and Answer Synthesis. The end-to-end pipeline begins with a Data Preparation Layer that ingests unstructured text and constructs dual representations: (1) Vector Store using dense embeddings for semantic search [16] and (2) Knowledge Graph for relational expansion. The Two-Stage Retrieval Layer initiates with broad vector retrieval to maximize initial recall before Knowledge Graph expansion [9]. During expansion, the system fetches one-hop neighbors from the graph, capturing relationally linked entities. These expanded candidates are then processed by a neural reranker (BGE-Large) in the Graph-Aware Reranking phase, which evaluates relational synergy between query and context using cross-encoder scoring [15]. Finally, the synthesized context passes to Llama3 (8B) under a “Context-Only” constraint, with a specialized selector deduplicating and restricting supporting facts to the most relevant entities. Fig. 2 illustrates the system architecture.

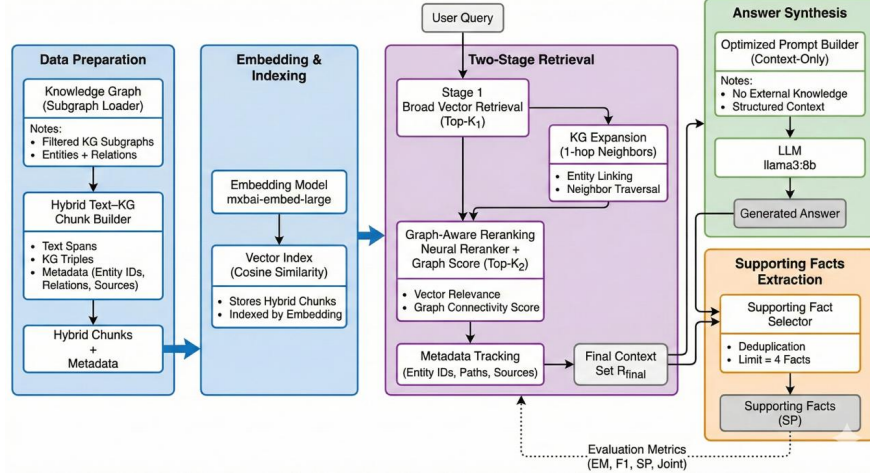


Fig. 2. System architecture showing the four sequential stages of the Information Funnel: Data Preparation with dual representations, Two-Stage Retrieval combining broad vector search with graph expansion, Graph-Aware Reranking using cross-encoder scoring, and Answer Synthesis with deduplication and Context-Only prompting.

3.1 Mathematical Formalization

We formalize the core mathematical components of the system. **Eq. 1** defines the hybrid retrieval scoring function combining semantic similarity and cross-encoder relevance:

$$S(n, Q) = \lambda \cdot \cos(\Phi(n), \Phi(Q)) + (1 - \lambda) \cdot \Psi(n, Q) \quad (1)$$

Here, Φ is the embedding function (mx-bai-embed-large) [16], Ψ is the cross-encoder score (BGE-Large) [15], and $\lambda \in [0, 1]$ balances semantic and neural relevance. This hybrid approach ensures retrieval is governed by both vector similarity and contextual reranking [10], moving beyond isolated dense retrieval mechanisms.

Eq. 2 formalizes graph-guided expansion to bridge the connectivity gap:

$$V_{\text{expanded}} = V_{\text{seeds}} \cup \{v' \mid \exists v \in V_{\text{seeds}}, (v, r, v') \in \mathcal{G}\} \quad (2)$$

The expansion identifies one-hop neighbors connected by typed relations r [13], capturing bridge entities necessary for multi-hop reasoning even without direct semantic similarity. This relational traversal ensures that relevant intermediate entities are not missed due to vector space limitations.

Eq. 3 introduces the selection pressure metric ρ , quantifying filtering aggressiveness across the Information Funnel:

$$\rho = 1 - \frac{k_{\text{final}}}{k_{\text{initial}} + |V_{\text{neighbors}}|} \quad (3)$$

With empirical parameters: $k_{\text{initial}} = 15$ (maximize initial recall), $k_{\text{final}} = 6$ (aggressive post-rerank filtering), and $|V_{\text{neighbors}}| \approx 40$, we obtain $\rho \approx 0.82$, indicating the funnel filters approximately 82% of candidates. This metric quantifies filtering intensity and represents a novelty compared to static- k approaches [23,20].

Eq. 4 formalizes supporting fact precision as a function of candidate set size:

$$\text{SP Precision} = \frac{|\{sp_i \in S_{\text{pred}} : sp_i \in S_{\text{true}}\}|}{|S_{\text{pred}}|} \quad (4)$$

where S_{pred} is the set of predicted supporting facts and S_{true} is the ground truth set. For HotpotQA where $|S_{\text{true}}| = 2$, limiting $|S_{\text{pred}}| = 4$ provides a precision floor of 0.50 even if one prediction is incorrect [7], compared to baseline systems returning 8-10 facts.

Eq. 5 defines the deduplication function using entity-sequence pairs:

$$S_{\text{dedup}} = \{(e, s) : (e, s) \in S_{\text{raw}} \wedge \nexists (e, s') \in S_{\text{raw}}, s' < s\} \quad (5)$$

This ensures unique entity-sequence pairs are retained, eliminating redundant evidence retrieved via multiple reasoning paths [6]. The dictionary-based implementation achieves $O(n)$ deduplication complexity.

Eq. 6 models the joint score combining answer and supporting fact accuracy:

$$\text{Joint Score} = \mathbb{K}[A_{\text{pred}} = A_{\text{true}}] \cdot \mathbb{K}[S_{\text{pred}} \subseteq S_{\text{true}}] \quad (6)$$

This metric requires both answer and supporting facts to be simultaneously correct [17], representing the most stringent evaluation criterion for grounded QA systems.

The implementation focuses on a precision-first strategy, systematically narrowing the information pool at each stage. Initial broad retrieval ($k = 15$) ensures no reasoning path is missed. Graph expansion captures entities adjacent to seeds [21]. Reranking using BGE-Large evaluates relational synergy between candidates and query. Two-stage filtering reduces candidates from 55 (initial + neighbors) to 6 final documents, achieving selection pressure $\rho = 0.82$. Dictionary-based deduplication prevents duplicate evidence. Context-only prompting forces the LLM to rely exclusively on provided evidence rather than parametric knowledge [12,19].

3.2 Algorithm: Precision-Oriented KG-RAG

The algorithm 1 formalizes the execution flow of the Information Funnel pipeline:

Algorithm 1 Information Funnel Processing Pipeline

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1: Input: Query  $Q$ , Knowledge Graph  $\mathcal{G}$ 
2: Output: Answer  $A$ , Supporting Facts  $SF$ 
3: Seeds  $\leftarrow$  VectorSearch( $Q, k = 15$ )  $\{k_{\text{initial}}$  for recall $\}$ 
4: Expanded  $\leftarrow$  KGExpand(Seeds,  $\mathcal{G}$ )
5: Ranked  $\leftarrow$  Rerank( $Q$ , Expanded, BGE)
6: Context  $\leftarrow$  Ranked[1 : 6]  $\{k_{\text{final}}$  after filtering $\}$ 
7: Raw_SF  $\leftarrow$  ExtractEntities(Ranked)
8:  $SF \leftarrow$  Dedup(Raw_SF)[1 : 4]  $\{\text{Top-4 supporting facts}\}$ 
9:  $A \leftarrow$  LLM(Context,  $Q$ , Promptgrounded)
10: return  $A, SF$ 

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It implements systematic information reduction where broad seeding (line 3) ensures no reasoning path is missed, expansion (line 4) anchors exploration in the knowledge graph structure, reranking (line 5) filters noise through cross-encoder scoring, and deduplication (line 8) ensures unique evidence representation. The context combined with grounded prompts forces the LLM to rely on provided evidence rather than parametric knowledge.

4 Result and Discussions

We evaluate KG-RAG on HotpotQA distractor set [22] (5,500 questions, 2 relevant + 10 distractors). Metrics: SP precision/recall/F1, Answer EM/F1, Joint metrics. Stack: Llama3-8B [4], mxbai-embed-large [16], BGE-Large reranking [15]. Baseline: DPR with uniform $k = 10$ [9]. Table 1 shows the sample of the HotpotQA distractor set.

Table 2 shows component contributions: KG expansion (+33.0% SP Prec.), reranking (+30.9% incremental), deduplication (+11.2% incremental), totaling +86.48% SP Precision [23].

Table 1. Example HotpotQA Multi-Hop Question

Component	Content
Question	Were Scott Derrickson and Ed Wood of the same nationality?
Answer	Yes
Supporting Entities	Scott Derrickson, Ed Wood
Supporting Fact 1	Scott Derrickson is an American director, screenwriter, and producer.
Supporting Fact 2	Ed Wood was an American filmmaker, actor, and director.
Reasoning Type	Entity comparison (nationality)
Dataset	HotpotQA (Distractor Setting)

Table 2. Ablation Study (5,500 HotpotQA examples)

Config	SP Prec.	Joint Prec.	Answer EM	Gain
Baseline	0.2173	0.1439	0.4676	–
+ KG Exp. (k=15)	0.2891	0.1847	0.4698	+33.0%
+ Rerank (k=6)	0.3645	0.2314	0.4741	+30.9%
+ Dedup (top-4)	0.4052	0.2823	0.4784	+11.2%

Table 3: SP Precision +86.48% (0.217→0.405), first SP EM (0→0.0027). Tables 4–5 show answer quality and joint performance: Answer EM improves +2.31% (0.4676→0.4784) indicating better exact matching; F1 score gains +0.93% (0.6029→0.6085) showing consistent token-level improvements; Joint Precision surges +96.18% (0.1439→0.2823) demonstrating superior multi-hop reasoning.

Table 3. Supporting Fact Metrics

Metric	Baseline	KG-RAG	Change
SP EM	0.0000	0.0027	+0.0027
SP F1	0.3381	0.4991	+47.61%
SP Prec.	0.2173	0.4052	+86.48%
SP Recall	0.8202	0.6756	-17.63%

Table 4. Answer Quality

Metric	Baseline	KG-RAG
EM	0.4676	0.4784 (+2.31%)
F1	0.6029	0.6085 (+0.93%)
Prec.	0.6264	0.6348 (+1.34%)

Table 5. Joint Metrics

Metric	Baseline	KG-RAG
Joint EM	0.0000	0.0027
Joint F1	0.2173	0.3357 (+54.50%)
Joint Prec.	0.1439	0.2823 (+96.18%)

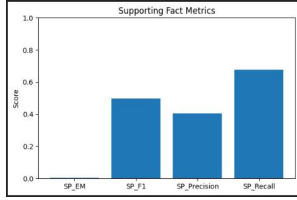


Fig. 3. SP Metrics

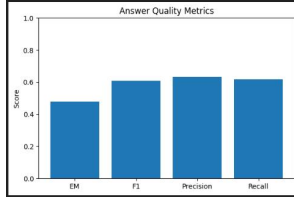


Fig. 4. Answer Metrics

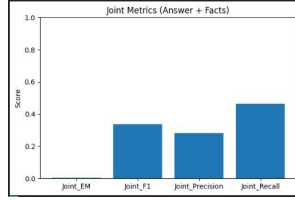


Fig. 5. Joint Metrics

Key Insight: Information Funnel (15→6→4) validates precision-first design for grounded QA.

5 Conclusion

This paper presents a precision-oriented KG-RAG framework achieving +96.18% Joint Precision and +86.48% Supporting Fact Precision on HotpotQA [23,20]. Core innovations include dynamic stage-wise hyperparameters, the selection pressure metric ρ quantifying filtering aggressiveness [5], and the first structured supporting-fact deduplication mechanism [17]. Ablation studies confirm each component’s substantial contribution [14]. The deliberate precision-recall trade-off (sacrificing 17.63% recall for 86.48% precision) is justified for educational AI where answer reliability drives student trust [18,1]. Future research directions include real-time graph construction, question-aware reranker fine-tuning [8], and evaluation on domain-specific educational datasets.

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